Oracle
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
Rectified gain @ k=24 (kcal/mol)
Predicted most stabilizing per domain
10
1
0
3.28 1.80 6.68 6.09
1.21 1.01 4.66 3.29 1.57 0.96 4.64 3.70 0.76 0.78 2.75 2.60 1.04 0.56 4.21 2.91 1.35 0.69 5.02 2.55
1.05 1.25 1.31 1.32 1.44 1.77 1.65 1.18 0.40 0.24 0.82 0.39
1.07 0.30 3.14 1.19 0.99 0.50 2.86 1.89
4.00 2.67 8.61 8.73
2.86 2.65 9.98 8.52 3.11 2.34 9.66 8.89 2.75 3.42 8.38 9.44 3.15 2.10 10.80 7.95 3.14 1.99 12.17 6.70
5.09 8.15 3.85 6.91 6.33 8.45 4.48 4.73 6.64 5.58 6.21 5.58
5.55 1.42 11.18 5.85 3.42 3.15 8.85 8.58
Top 24 per domain
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta
Δ hydrophobicity ≤ 0 & relative ASA ≤ 0.5
Δ hydrophobicity > 0 & relative ASA ≤ 0.5
Δ hydrophobicity ≤ 0 & relative ASA > 0.5
Δ hydrophobicity > 0 & relative ASA > 0.5

Top 24 per Homology-Restricted MegaScale test domain (33 domains with the feature, 35,295 single mutants); gain = sum of max(ddG, 0) over the picks predicted stabilizing (> 0);
boxes: interquartile range with darker colored median line; dots: domains; dark bars & annotations: mean over domains; dashed lines: oracle mean.